Physics 468/668: Statistical Physics **Spring 2019**

Lecture 8: The Canonical Ensemble

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8.1 Density of States

In most physical cases the energy levels accessible to a system is degenerate, that is, one has a group of states g_i, belonging to the same energy value E_i. In such cases, it is more convenient to write the partition function as

$$Q_N(V,T) = \sum_i g_i \exp(-\beta E_i) \tag{8.1}$$

The corresponding expression for P_i would be

$$P_i = \frac{g_i \exp(-\beta E_i)}{\sum_i g_i \exp(-\beta E_i)} \tag{8.2}$$

Clearly, the g_i states are all equally likely to occur. As a result, the probability of a system having energy E_i becomes proportional to the multiplicity g_i of this level.

In view of the largeness of the number of particles constituting a given system and the largeness of the volume to which these particles are confined, the consecutive energy values E_i of the system are, in general, very close to one another. Accordingly, there lie, within any reasonable interval of energy $(E, E+dE)$, a very large number of energy levels.

Denoting it by $g(E)dE$, where $g(E)$ denotes the density of states around energy value E, we have

$$P(E)dE \propto \exp(-\beta E) g(E) dE \tag{8.3}$$

After normalization,

$$P(E)dE = \frac{\exp(-\beta E) g(E) dE}{\int_0^\infty \exp(-\beta E) g(E) dE} \tag{8.4}$$

The denominator here is yet another expression for the partition function,

$$Q_N(V,T) = \int_0^\infty \exp(-\beta E) g(E) dE \tag{8.5}$$

With $\beta > 0$, the partition function $Q(\beta)$ is just the Laplace transform of $g(E)$. We may therefore, write $g(E)$ as the inverse Laplace transform of $Q(\beta)$

$$\begin{aligned} g(E) &= \frac{1}{2\pi i} \int_{\beta'-i\infty}^{\beta'+i\infty} \exp(\beta E) Q(\beta) d\beta \ (\beta' > 0) \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp[(\beta' + i\beta'')E] Q(\beta' + i\beta'') d\beta'' \end{aligned} \tag{8.6}$$

Accordingly, the averaged value could be written as,

$$< f >= \sum_i f_i P_i = \frac{\sum_i f(E_i) g_i \exp(-\beta E_i)}{\sum_i g_i \exp(-\beta E_i)} \rightarrow \frac{\int_0^\infty f(E) \exp(-\beta E) g(E) dE}{\int_0^\infty \exp(-\beta E) g(E) dE} \tag{8.7}$$

8.2 The classical systems

The theory developed in the preceding sections is of general applicability. We recall the concepts in ensemble theory, the averaged quantity $< f >$ is,

$$< f >= \frac{\int f(q,p)\rho(q,p) d^{3N}q d^{3N}p}{\int \rho(q,p) d^{3N}q d^{3N}p} \tag{8.8}$$

For canonical system, it follows,

$$\rho(q,p) \propto \exp[-\beta H(q,p)] \tag{8.9}$$

The expression for $< f >$ then take the form

$$< f >= \frac{\int f(q,p) \exp(\beta H) d\omega}{\int \exp(\beta H) d\omega} \tag{8.10}$$

The denominator of this expression is directly related to the partition function of the system. However, we must take into account the relationship between a volume element in the phase space, and the corresponding number of distinct quantum states of the system. The relationship is

$$\frac{d\omega}{N! h^{3N}} \tag{8.11}$$

The appropriate expression for the partition function would be,

$$Q_N(V,T) = \frac{1}{N! h^{3N}} \int \exp(-\beta H(q,p)) d\omega \tag{8.12}$$

We consider an ideal gas with N identical molecules, confined to V and in equilibrium at T. Since there are no intermolecular interactions to be taken into account, the energy is

$$H(q,p) = \sum_{i=1}^{N} (p_i^2/2m) \tag{8.13}$$

The partition function of the system would be then,

$$Q_N(V,T) = \frac{1}{N! h^{3N}} \int \exp\left[-(\beta/2m) \sum_i p_i^2 \right] \prod_{i=1}^{N} (d^3 q_i d^3 p_i) \tag{8.14}$$

Integration over the space gives V^N. Integrations over the momentum space are also trivial,

$$Q_N(V,T) = \frac{V^N}{N! h^{3N}} \left[\int_0^\infty \exp(-p^2/2mkT)(4\pi p^2 dp) \right]^N = \frac{1}{N!} \left[\frac{V}{h^3} (2\pi mkT)^{3/2} \right]^N \tag{8.15}$$

The Helmholtz energy is then given by

$$A(N,V,T) = -kT \ln Q_N(V,T) = NkT\left[\ln\left(\frac{N}{V}(\frac{h^2}{2\pi mkT})^{3/2}\right) - 1\right] \tag{8.16}$$

With knowing A, we can get the rest properties,

$$\mu = \left(\frac{\partial A}{\partial N}\right)_{V,T} = kT \ln\left[\frac{N}{V}\left(\frac{h^2}{2\pi mkT}\right)^{3/2}\right] \tag{8.17}$$

$$P = \left(\frac{\partial A}{\partial V}\right)_{N,T} = \frac{NkT}{V} \tag{8.18}$$

$$S = \left(\frac{\partial A}{\partial T}\right)_{N,V} = Nk\left[\ln\left(\frac{N}{V}(\frac{h^2}{2\pi mkT})^{3/2}\right) + \frac{5}{2}\right] \tag{8.19}$$

The results are similar to what we derived from thermodynamics.

At this stage, we may write

$$Q_N(V,T) = \frac{1}{N!}[Q_1(V,T)]^N \tag{8.20}$$

Going back to the ideal gas, we could as well start with the density of states $g(E)$.

$$g(E) = \frac{\partial \sum}{\partial E} \approx \frac{1}{N!}\left(\frac{V}{h^3}\right)^N \frac{(2\pi m)^{3N/2}}{[(3N/2-1)]!} E^{3N/2-1} \tag{8.21}$$

Substituting this into eq(*), we obtain

$$Q_N(\beta) = \frac{1}{N!}\left(\frac{V}{h^3}\right)^N \left(\frac{2\pi m}{\beta}\right)^{3N/2} \tag{8.22}$$

If we start with the single particle, we will get

$$\alpha(\epsilon) = \frac{2\pi V}{h^3}(2m)^{3/2}\epsilon^{1/2} \tag{8.23}$$

And the corresponding single-particle partition function

$$Q_1(\beta) = \int_0^\infty \exp(-\beta\epsilon)\alpha(\epsilon)d\epsilon = \frac{V}{h^3}\left(\frac{2\pi m}{\beta}\right)^{3/2} \tag{8.24}$$

How to calculate $g(E)$ from Q? Assuming that Q is known, we have

$$g(E) = \frac{V^N}{N!}\left(\frac{2\pi m}{h^2}\right)^{3N/2} \frac{1}{2\pi i}\int_{\beta'+i\infty}^{\beta'-i\infty} \frac{\exp(\beta E)}{\beta^{3N/2}} d\beta \tag{8.25}$$

Noting that for all positive n,

$$\frac{1}{2\pi i}\int_{s'-i\infty}^{s'+i\infty} \frac{\exp(sx)}{s^{n+1}} ds = \begin{cases} \frac{x^n}{n!} & \text{if } x \geq 0 \\ 0 & \text{if } x \leq 0 \end{cases} \tag{8.26}$$

Therefore

$$g(E) = \begin{cases} \frac{V^N}{N!}\left(\frac{2\pi m}{h^2}\right)^{3N/2} \frac{E^{(3N/2)-1}}{(3N/2-1)!} & \text{if } E \geq 0 \\ 0 & \text{if } E < 0 \end{cases} \tag{8.27}$$